CAS4160 Homework 2: Policy Gradients

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1 Introduction

The goal of this assignment is to help you understand **Policy Gradient Methods**, including variance reduction tricks, such as reward-to-go, neural network baselines, and Generalized Advantage Estimator (GAE). Finally, you will implement Proximal Policy Optimization (PPO), a widely used on-policy RL algorithm that works very well in practice. Here is the link to this homework report template in Overleaf.

2 Review

2.1 Policy Gradient

Recall that the reinforcement learning objective is to learn θ^* that maximizes the objective function:

$$J(\theta) = \mathbb{E}_{\tau\sim\pi_\theta(\tau)}[r(\tau)], \tag{1}$$

where each rollout τ is of length T, as follows:

$$\pi_\theta(\tau) = p\left(\mathbf{s}_0, \mathbf{a}_0, \ldots, \mathbf{s}_{T-1}, \mathbf{a}_{T-1}, \mathbf{s}_T\right) = p\left(\mathbf{s}_0\right) \prod_{t=0}^{T-1} \pi_\theta\left(\mathbf{a}_t \mid \mathbf{s}_t\right) p\left(\mathbf{s}_{t+1} \mid \mathbf{s}_t, \mathbf{a}_t\right),$$

and

$$r(\tau) = r\left(\mathbf{s}_0, \mathbf{a}_0, \ldots, \mathbf{s}_{T-1}, \mathbf{a}_{T-1}\right) = \sum_{t=0}^{T-1} r\left(\mathbf{s}_t, \mathbf{a}_t\right).$$

Note: In this assignment, t starts from 0, not from 1, which is easier to implement.

The policy gradient approach is to directly take the gradient of this objective:

$$\begin{aligned}
\nabla_\theta J(\theta) &= \nabla_\theta \mathbb{E}_{\tau\sim\pi_\theta(\tau)}[r(\tau)] \\
&= \nabla_\theta \int \pi_\theta(\tau) r(\tau) d\tau \\
&= \int \pi_\theta(\tau) \nabla_\theta \log \pi_\theta(\tau) r(\tau) d\tau \qquad (\because \nabla_\theta \pi_\theta(\tau) = \pi_\theta(\tau) \nabla_\theta \log \pi_\theta(\tau)) \\
&= \mathbb{E}_{\tau\sim\pi_\theta(\tau)}\left[\nabla_\theta \log \pi_\theta(\tau) r(\tau)\right].
\end{aligned}$$

In practice, the expectation over trajectories τ can be approximated from a batch of N sampled trajectories:

$$\begin{aligned}
\nabla_\theta J(\theta) &\approx \frac{1}{N} \sum_{i=1}^{N} \nabla_\theta \log \pi_\theta\left(\tau_i\right) r\left(\tau_i\right) \\
&= \frac{1}{N} \sum_{i=1}^{N} \left(\sum_{t=0}^{T-1} \nabla_\theta \log \pi_\theta\left(\mathbf{a}_{it} \mid \mathbf{s}_{it}\right)\right) \left(\sum_{t=0}^{T-1} r\left(\mathbf{s}_{it}, \mathbf{a}_{it}\right)\right).
\end{aligned} \tag{2}$$

Here, we see that the policy π_θ is a probability distribution over the action space, conditioned on the state. In the agent-environment loop, the agent samples an action $\mathbf{a}_t$ from $\pi_\theta(\cdot \mid \mathbf{s}_t)$ and the environment responds with a reward $r(\mathbf{s}_t, \mathbf{a}_t)$.

2.2 Variance Reduction

2.2.1 Reward-to-go

One way to reduce the variance of the policy gradient is to exploit causality: the notion that the policy cannot affect the rewards in the past. This yields the following modified objective, where the sum of rewards here does not include the rewards achieved prior to the time step at which the policy is being queried. This sum of rewards is a sample estimate of the Q function, and is referred to as the "reward-to-go":

$$\nabla_\theta J(\theta) \approx \frac{1}{N} \sum_{i=1}^{N} \sum_{t=0}^{T-1} \nabla_\theta \log \pi_\theta(\mathbf{a}_{it} \mid \mathbf{s}_{it}) \underbrace{\left(\sum_{t'=t}^{T-1} r(\mathbf{s}_{it'}, \mathbf{a}_{it'}) \right)}_{\text{reward-to-go}}. \tag{3}$$

2.2.2 Discounting

Multiplying a discount factor γ to the rewards can be interpreted as encouraging the agent to focus more on the rewards that are closer in time, and less on the rewards that are further in the future. This can also be thought of as a means for reducing variance as there can be more variance and uncertainty when considering faraway futures.

We can apply the discount on the rewards from full trajectory in Equation (2):

$$\nabla_\theta J(\theta) \approx \frac{1}{N} \sum_{i=1}^{N} \left(\sum_{t=0}^{T-1} \nabla_\theta \log \pi_\theta(\mathbf{a}_{it} \mid \mathbf{s}_{it}) \right) \left(\sum_{t'=0}^{T-1} \gamma^{t'-1} r(\mathbf{s}_{it'}, \mathbf{a}_{it'}) \right). \tag{4}$$

We can also apply the discount on the "reward-to-go" in Equation (3):

$$\nabla_\theta J(\theta) \approx \frac{1}{N} \sum_{i=1}^{N} \sum_{t=0}^{T-1} \nabla_\theta \log \pi_\theta(\mathbf{a}_{it} \mid \mathbf{s}_{it}) \left(\sum_{t'=t}^{T-1} \gamma^{t'-t} r(\mathbf{s}_{it'}, \mathbf{a}_{it'}) \right). \tag{5}$$

2.2.3 Baseline

Another variance reduction method is to subtract a baseline (that is only conditioned on the current state $\mathbf{s}_t$) from the sum of rewards:

$$\nabla_\theta J(\theta) = \mathbb{E}_{\tau \sim \pi_\theta(\tau)} \left[\sum_{t=0}^{T-1} \nabla_\theta \log \pi_\theta(\mathbf{a}_t \mid \mathbf{s}_t) [r(\tau) - b(\mathbf{s}_t)] \right]. \tag{6}$$

This leaves the policy gradient unbiased because

$$\begin{aligned} \mathbb{E}_{\tau \sim \pi_\theta(\tau)}[\nabla_\theta \log \pi_\theta(\mathbf{a}_t \mid \mathbf{s}_t) b(\mathbf{s}_t)] &= \mathbb{E}_{\tau \sim \pi_\theta(\tau)}[\nabla_\theta \log \pi_\theta(\mathbf{a}_t \mid \mathbf{s}_t)] \mathbb{E}_{\tau \sim \pi_\theta(\tau)}[b(\mathbf{s}_t)] \\ &= 0 \cdot \mathbb{E}_{\tau \sim \pi_\theta(\tau)}[b(\mathbf{s}_t)] = 0. \end{aligned}$$

In this assignment, we will implement a value function $V_\phi^\pi(\mathbf{s}_t)$, which acts as a state-dependent baseline. This value function will be trained to approximate the sum of future rewards starting from a particular state:

$$V_\phi^\pi(\mathbf{s}_t) \approx \sum_{t'=t}^{T-1} \mathbb{E}_{\tau \sim \pi_\theta(\tau)} \left[r(\mathbf{s}_{t'}, \mathbf{a}_{t'}) \mid \mathbf{s}_t \right]. \tag{7}$$

Then, the approximate policy gradient now looks like this:

$$\nabla_\theta J(\theta) \approx \frac{1}{N} \sum_{i=1}^{N} \sum_{t=0}^{T-1} \nabla_\theta \log \pi_\theta \left(\mathbf{a}_{it} \mid \mathbf{s}_{it} \right) \left(\underbrace{\left(\sum_{t'=t}^{T-1} \gamma^{t'-t} r \left(\mathbf{s}_{it'}, \mathbf{a}_{it'} \right) \right)}_{\text{reward-to-go}} - \underbrace{V_\phi^\pi \left(\mathbf{s}_{it} \right)}_{\text{baseline}} \right). \tag{8}$$

2.2.4 Generalized Advantage Estimator (GAE)

The quantity $\left(\sum_{t'=t}^{T-1} \gamma^{t'-t} r \left(\mathbf{s}_{t'}, \mathbf{a}_{t'} \right) \right) - V_\phi^\pi \left(\mathbf{s}_t \right)$ from the previous policy gradient expression (removing the index i for clarity) can be interpreted as an estimate of the advantage function:

$$A^\pi \left(\mathbf{s}_t, \mathbf{a}_t \right) = Q^\pi \left(\mathbf{s}_t, \mathbf{a}_t \right) - V^\pi \left(\mathbf{s}_t \right), \tag{9}$$

where $Q^\pi \left(\mathbf{s}_t, \mathbf{a}_t \right)$ is estimated using Monte Carlo returns and $V^\pi \left(\mathbf{s}_t \right)$ is estimated using the learned value function V_ϕ^π. We can further reduce variance by also using V_ϕ^π in place of the Monte Carlo returns to estimate the advantage function as:

$$A^\pi \left(\mathbf{s}_t, \mathbf{a}_t \right) \approx \delta_t = r \left(\mathbf{s}_t, \mathbf{a}_t \right) + \gamma V_\phi^\pi \left(\mathbf{s}_{t+1} \right) - V_\phi^\pi \left(\mathbf{s}_t \right), \tag{10}$$

with the edge case $\delta_{T-1} = r \left(\mathbf{s}_{T-1}, \mathbf{a}_{T-1} \right) - V_\phi^\pi \left(\mathbf{s}_{T-1} \right)$. However, this comes at the cost of introducing bias to our policy gradient estimate, due to modeling errors in V_ϕ^π. We can instead use a combination of n-step Monte Carlo returns and V_ϕ^π to estimate the advantage function as:

$$A_n^\pi \left(\mathbf{s}_t, \mathbf{a}_t \right) = \sum_{t'=t}^{t+n} \gamma^{t'-t} r \left(\mathbf{s}_{t'}, \mathbf{a}_{t'} \right) + \gamma^n V_\phi^\pi \left(\mathbf{s}_{t+n+1} \right) - V_\phi^\pi \left(\mathbf{s}_t \right). \tag{11}$$

Increasing n incorporates the Monte Carlo returns more heavily in the advantage estimate, which lowers bias and increases variance, while decreasing n does the opposite. Note that $n = T - t - 1$ recovers the unbiased but higher variance Monte Carlo advantage estimate used in (13), while $n = 0$ recovers the lower variance but higher bias advantage estimate δ_t.

We can combine multiple n-step advantage estimates as an exponentially weighted average, which is known as the generalized advantage estimator (GAE). Let $\lambda \in [0, 1]$. Then, we define:

$$A_{GAE}^\pi \left(\mathbf{s}_t, \mathbf{a}_t \right) = \frac{1 - \lambda^{T-t-1}}{1 - \lambda} \sum_{n=1}^{T-t-1} \lambda^{n-1} A_n^\pi \left(\mathbf{s}_t, \mathbf{a}_t \right), \tag{12}$$

where $\frac{1-\lambda^{T-t-1}}{1-\lambda}$ is a normalizing constant. Note that a higher λ emphasizes advantage estimates with higher values of n, and a lower λ does the opposite. Thus, λ serves as a control for the bias-variance tradeoff, where increasing λ decreases bias and increases variance. In the infinite horizon case ($T = \infty$), we can show:

$$\begin{aligned} A_{GAE}^\pi \left(\mathbf{s}_t, \mathbf{a}_t \right) &= \frac{1}{1 - \lambda} \sum_{n=1}^{\infty} \lambda^{n-1} A_n^\pi \left(\mathbf{s}_t, \mathbf{a}_t \right) \\ &= \sum_{t'=t}^{\infty} (\gamma\lambda)^{t'-t} \delta_{t'}, \end{aligned}$$

where we have omitted the derivation for brevity (see the GAE paper for details). In the finite horizon case, we can write:

$$A_{GAE}^\pi \left(\mathbf{s}_t, \mathbf{a}_t \right) = \sum_{t'=t}^{T-1} (\gamma\lambda)^{t'-t} \delta_{t'}, \tag{13}$$

which serves as a way we can efficiently implement the generalized advantage estimator, since we can recursively compute:

$$A^{\pi}_{GAE}\left(\mathbf{s}_t, \mathbf{a}_t\right) = \delta_t + \gamma\lambda A^{\pi}_{GAE}\left(\mathbf{s}_{t+1}, \mathbf{a}_{t+1}\right) \tag{14}$$

3 Code Structure Overview

The training begins with the script `run_hw2.py`. This script contains the function `run_training_loop`, where the training, evaluation, and logging happens.

The agent, also defined in `run_hw2.py`, is an instance of `PGAgent` in `cas4160/agents/pg_agent.py`. The `PGAgent` class has two main networks as its variables:

- **actor** network: `MLPPolicyPG` in `cas4160/networks/policies.py`, takes as input observations and outputs actions.
- **critic** network: `ValueCritic` in `cas4160/networks/critics.py`, also takes as input observations but outputs value estimates. You will need to implement this in Section 5.

The training iteration begins with sampling trajectories using `sample_trajectories()` defined in `infrastructure/utils.py` and updating the agent via `PGAgent.update()` in `agents/pg_agent.py`.

In this `update()` procedure, `PGAgent` first processes rewards into Q-values (`PGAgent._calculate_q_vals()`) and then, estimates advantages (`PGAgent._estimate_advantage()`). Then, the actor and the critic are updated via `MLPPolicyPG.update()` in `cas4160/networks/policies.py` and `ValueCritic.update()` in `cas4160/networks/cr` respectively. If the `--use_ppo` flag is set, the `MLPPolicyPG.ppo_update()` method is used instead of the standard `MLPPolicyPG.update()` method for the actor. You will work on the `MLPPolicyPG.ppo_update()` method in Section 7.

4 Policy Gradients

4.1 Implementation

You will be implementing two different return estimators within `pg_agent.py`. Note that these differ only by the starting point of the summation.

The first ("Case 1" within `PGAgent._calculate_q_vals()`) uses the discounted cumulative return of the full trajectory and corresponds to the "vanilla" form of the policy gradient (Equation (2)):

$$r\left(\tau_i\right) = \sum_{t'=0}^{T-1} \gamma^{t'} r\left(\mathbf{s}_{it'}, \mathbf{a}_{it'}\right). \tag{15}$$

The second ("Case 2") uses the "reward-to-go" formulation from Equation (3):

$$r\left(\tau_i\right) = \sum_{t'=t}^{T-1} \gamma^{t'-t} r\left(\mathbf{s}_{it'}, \mathbf{a}_{it'}\right). \tag{16}$$

Implement these return estimators as well as the remaining sections marked TODO in the code. For the small-scale experiments, you may skip the parts that are run only if **nn_baseline is True**; we will return to baselines (`MLPPolicyPG.update()` and `PGAgent._estimate_advantage()`) in Section 5.

4.2 Experiments

Experiment 1 (CartPole). Run multiple experiments with the PG algorithm on the discrete `CartPole-v0` environment, using the following commands:

```
python cas4160/scripts/run_hw2.py --env_name CartPole-v0 -n 100 -b 1000 \
    --exp_name cartpole

python cas4160/scripts/run_hw2.py --env_name CartPole-v0 -n 100 -b 1000 \
    -rtg --exp_name cartpole_rtg

python cas4160/scripts/run_hw2.py --env_name CartPole-v0 -n 100 -b 1000 \
    -na --exp_name cartpole_na

python cas4160/scripts/run_hw2.py --env_name CartPole-v0 -n 100 -b 1000 \
    -rtg -na --exp_name cartpole_rtg_na

python cas4160/scripts/run_hw2.py --env_name CartPole-v0 -n 100 -b 4000 \
    --exp_name cartpole_lb

python cas4160/scripts/run_hw2.py --env_name CartPole-v0 -n 100 -b 4000 \
    -rtg --exp_name cartpole_lb_rtg

python cas4160/scripts/run_hw2.py --env_name CartPole-v0 -n 100 -b 4000 \
    -na --exp_name cartpole_lb_na

python cas4160/scripts/run_hw2.py --env_name CartPole-v0 -n 100 -b 4000 \
    -rtg -na --exp_name cartpole_lb_rtg_na
```

Here, each argument specifies:

- `-n` : Number of iterations.
- `-b` : Batch size (number of state-action pairs sampled while acting according to the current policy at each iteration).
- -rtg : Flag: if present, sets `reward_to_go=True`. Otherwise, `reward_to_go=False` by default.
- -na : Flag: if present, sets `normalize_advantages=True`, normalizing the advantages to have a mean of zero and standard deviation of one within a batch.
- –exp_name : Name for experiment, which goes into the name for the data logging directory. If your run succeeds, you will be able to find your tensorboard log data in `hw2_starter_code/data/q2_pg_[--exp_name]_[--env_name]_[current_time]/`.

Various other command line arguments will allow you to set batch size, learning rate, network architecture, and more. You can find list of arguments in `cas4160/scripts/run_hw2.py:main()`.

Note: To generate videos of the policy rollouts, add the flag "`--video_log_freq 10`". This will log the evaluation rollout for every 10 iterations. You can watch the video in the "`Images`" tab on tensorboard.

Deliverables for report:

- Create two graphs:
- In the first graph, compare the learning curves (average return vs. number of environment steps) for the experiments prefixed with `cartpole` (the small batch experiments).
- In the second graph, compare the learning curves for the experiments prefixed with `cartpole_lb` (the large batch experiments).

Note: For all plots in this assignment, the x-axis should be number of environment steps, logged as Train_EnvstepsSoFar (not number of policy gradient iterations).

Note: You can use the example helper script (`cas4160/scripts/parse_tensorboard.py`) to parse the data from the tensorboard logs and plot the figure. Here's an example usage that saves the figure as `output_plot.png`:

```
python cas4160/scripts/parse_tensorboard.py \
  --input_log_files data/[your_log_folder_1] data/[your_log_folder_2] \
  --human_readable_names "Vanilla" "Reward to go" \
  --data_key "Eval_AverageReturn" \
  --title "Your Title Here" \
  --x_label_name "Train Environment Steps" \
  --y_label_name "Eval Return" \
  --output_file "output_plot.png"
```

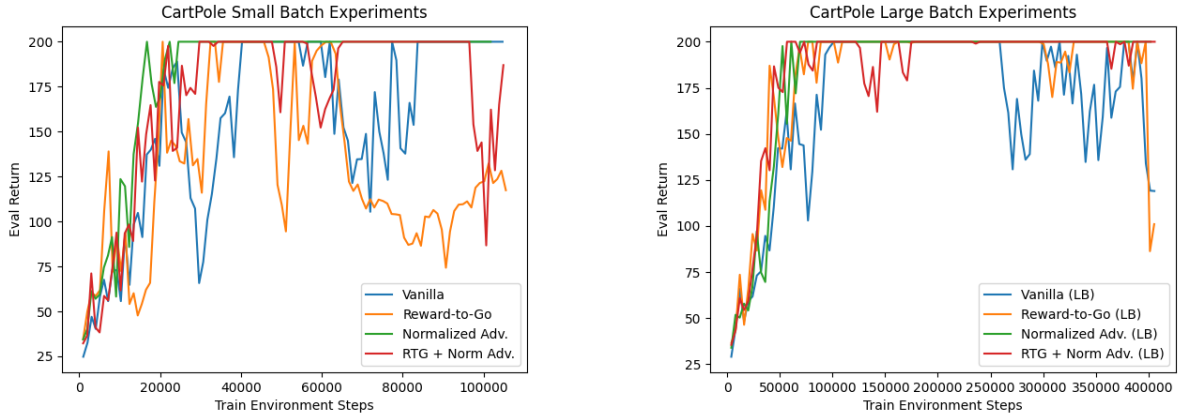
Note that if you add `--plot_mean_std` as the argument, you can plot the standard deviation as a shaded area across the `input_log_files` you specified. You can modify this parsing script for better visualization as you need.

What to Expect:

- The best configuration of CartPole in both the large and small batch cases should converge to a maximum score of 200. In addition, each run takes about 5 minutes without video logging.

4.3 Policy Gradient Result

4.3.1 Plot (Eval Average Return with batch size 1000 and 4000):



(a) Small Batch (batch size = 1000)

(b) Large Batch (batch size = 4000)

Figure 1: Eval Average Return curves for CartPole-v0. Left: small batch experiments, Right: large batch experiments. All experiments use 100 iterations.

In the case of small batches (batch size 1000), it was observed that applying reward-to-go and advantage normalization led to significant improvements in both learning speed and final performance. In particular, the combination of RTG + Normalized Advantage converged the fastest. The reward-to-go at timestep t is defined as

$$\mathbb{E}_{\tau\sim\pi_\theta(\tau)} \sum_{t=1}^{T} \nabla_\theta \log \pi_\theta(\mathbf{a}_t \mid \mathbf{s}_t) \left(\sum_{t'=t}^{T} r(\mathbf{s}_{t'}, \mathbf{a}_{t'}) \right) \tag{17}$$

In the case of large batches (batch size 4000), the overall performance was more stable and higher compared to small batches. It was observed that as the batch size increased, the variance of the gradients decreased, resulting in smoother learning progress. We can acknowledge this relationship using the below equeation.

$$\nabla_\theta J(\theta) \approx \frac{1}{B} \sum_{i=1}^{B} \nabla_\theta \log \pi_\theta(a_i \mid s_i) A_i \tag{18}$$

whose variance scales as $O(1/B)$, where B is the batch size.

In conclusion, it was confirmed that in Policy Gradient algorithms, reward-to-go, advantage normalization, and a sufficiently large batch size play a very important role in learning performance.

4.3.2 Questions:

Answer the following questions briefly:

- Which value estimator has better performance without advantage normalization: the sum of trajectory rewards (REINFORCE) or the one using reward-to-go?
  Answer: Reward-to-go

When advantage normalization was not used, the estimator using reward-to-go showed better performance than the REINFORCE using the sum of the entire trajectory rewards. Since reward-to-go considers only the future rewards after time t, the variance is greatly reduced, making the learning more stable and allowing it to converge faster.

- Did advantage normalization help?
  Answer:

  Advantage normalization greatly helped the learning performance. In the experiments where normalization was applied, the learning curve was smoother and more stable, and the final Eval Average Return was also higher. This is because it reduces the gradient variance caused by differences in the scale of the advantage, and the variance reduction effect was particularly pronounced when the batch size was small.

- Did the batch size make an impact?
  Answer:

  Batch size had a significant impact on learning. In the case of the large batch (batch size = 4000), the gradient variance was reduced compared to the small batch (batch size = 1000), resulting in a more stable learning curve, and the final performance was more consistent. This is because the variance of the gradient estimate decreases as the batch size increases. In the large batch experiments, it was observed that the effects of reward-to-go and advantage normalization were manifested more stably than in the small batch.

- Provide the exact command line configurations you used to run your experiments, including any parameters changed from their defaults.
  Answer:

- Small batch (batch size = 1000)
  - Vanilla: `python cas4160/scripts/run_hw2.py --env_name CartPole-v0 -n 100 -b 1000 --exp_name cartpole`
  - Reward-to-Go: `python cas4160/scripts/run_hw2.py --env_name CartPole-v0 -n 100 -b 1000 -rtg --exp_name cartpole_rtg`
  - Normalized Adv.: `python cas4160/scripts/run_hw2.py --env_name CartPole-v0 -n 100 -b 1000 -na --exp_name cartpole_na`
  - RTG + Normalized Adv.: `python cas4160/scripts/run_hw2.py --env_name CartPole-v0 -n 100 -b 1000 -rtg -na --exp_name cartpole_rtg_na`
- Large batch (batch size = 4000)
  - Vanilla: `python cas4160/scripts/run_hw2.py --env_name CartPole-v0 -n 100 -b 4000 --exp_name cartpole_lb`
  - Reward-to-Go: `python cas4160/scripts/run_hw2.py --env_name CartPole-v0 -n 100 -b 4000 -rtg --exp_name cartpole_lb_rtg`
  - Normalized Adv.: `python cas4160/scripts/run_hw2.py --env_name CartPole-v0 -n 100 -b 4000 -na --exp_name cartpole_lb_na`
  - RTG + Normalized Adv.: `python cas4160/scripts/run_hw2.py --env_name CartPole-v0 -n 100 -b 4000 -rtg -na --exp_name cartpole_lb_rtg_na`

  (Key parameters changed from the default values: -n 100, -b 1000 or 4000, -rtg, -na)

5 Using a Neural Network Baseline

5.1 Implementation

You will now implement a value function as a state-dependent neural network baseline. This will require filling in some TODO sections skipped in Section 4. In particular:

- This neural network will be trained in the update method of `MLPPolicyPG` along with the policy gradient update.
- In `PGAgent._estimate_advantage()`, the predictions of this network will be subtracted from the reward-to-go to yield an estimate of the advantage. This implements $\left(\sum_{t'=t}^{T-1} \gamma^{t'-t} r\left(\mathbf{s}_{it'}, \mathbf{a}_{it'}\right)\right) - V_\phi^\pi\left(\mathbf{s}_{it}\right)$.
- We will train the baseline network for multiple gradient steps for each policy update, determined by the parameter `baseline_gradient_steps`.

5.2 Experiments

Experiment 2 (HalfCheetah). Next, you will use your baselined policy gradient implementation to learn a controller for `HalfCheetah-v4`.

Run the following commands:

```
# No baseline
python cas4160/scripts/run_hw2.py --env_name HalfCheetah-v4 \
    -n 100 -b 5000 -rtg --discount 0.95 -lr 0.01 \
    --exp_name cheetah
# Baseline
python cas4160/scripts/run_hw2.py --env_name HalfCheetah-v4 \
    -n 100 -b 5000 -rtg --discount 0.95 -lr 0.01 \
    --use_baseline -blr 0.01 -bgs 5 --exp_name cheetah_baseline
```

You might notice that we omitted `-na` (normalize advantages). That's because in reality, advantage normalization is a very powerful trick, and eliminates the need for a baseline in most of the simple environments.

Deliverables:

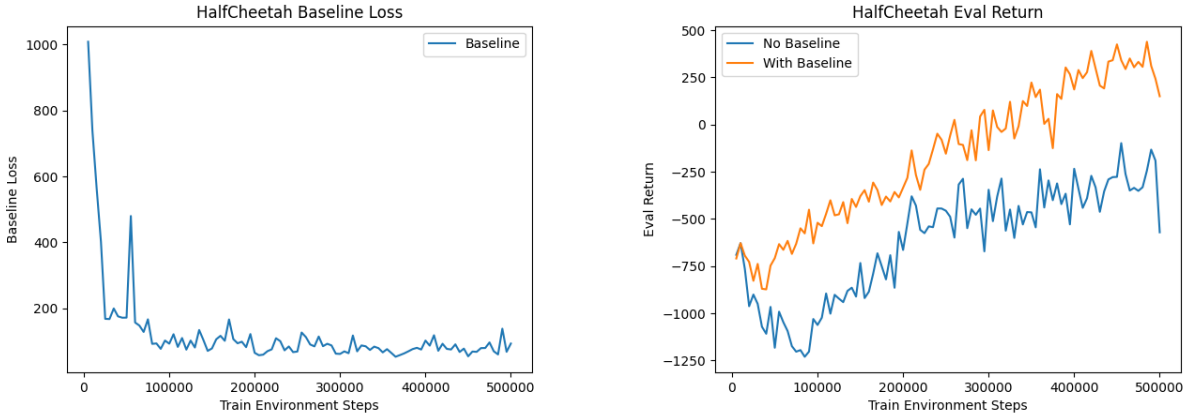
- Plot a learning curve for the baseline loss.
- Plot a learning curve for the eval return.
- Run another experiment with a decreased number of baseline gradient steps (`-bgs`) and/or baseline learning rate (`-blr`). How does this affect (a) the baseline learning curve and (b) the performance of the policy?
- (Optional) Add `-na` back to see how much it improves things. Also, set `--video_log_freq 10`, then open TensorBoard and go to the "Images" tab to see some videos of your HalfCheetah walking along!

What to Expect:

- You should expect to achieve an average return over 300 for the baseline version. In addition, each run takes about 10 minutes without video logging.

5.3 Neural Network Baseline Result

5.3.1 Plot (Baseline Loss, Eval Average Return):

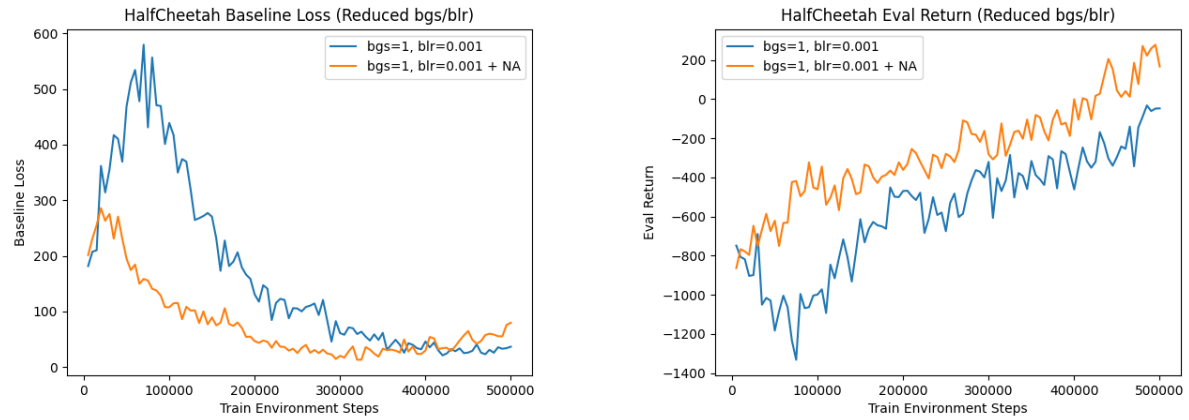


(a) Baseline Loss

(b) Eval Average Return

Figure 2: HalfCheetah-v4: Baseline Loss (left) and Eval Average Return (right) curves when using neural network baseline.

5.3.2 Plot (Reduced Baseline Gradient Steps, Learning Rate):



(a) Baseline Loss (bgs=1, blr=0.001)

(b) Eval Average Return (bgs=1, blr=0.001)

Figure 3: HalfCheetah-v4: Effect of reducing baseline gradient steps (`-bgs`) and learning rate (`-blr`). Left: Baseline Loss, Right: Eval Average Return (with and without advantage normalization).

5.3.3 Questions:

Answer the following questions briefly:

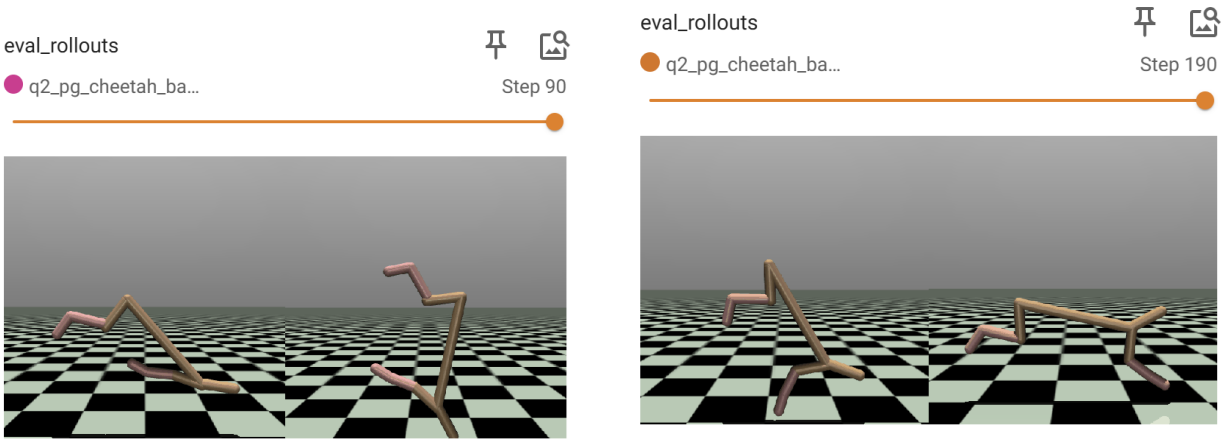
- Run another experiment with a decreased number of baseline gradient steps (`-bgs`) and/or baseline learning rate (`-blr`). How does this affect (a) the baseline learning curve and (b) the performance of

the policy?
Answer:

We conducted experiments in which the baseline gradient steps (`-bgs`) and baseline learning rate (`-blr`) were reduced (`-bgs=1`, `-blr=0.001`).

(a) **Baseline learning curve**: Reducing the baseline gradient steps and learning rate prevented the baseline network from being sufficiently trained, causing the loss curve to converge to a higher value and more unstably. In particular, when advantage normalization was not applied together, the loss fluctuated significantly, and overall the baseline tended to converge slowly or exhibit underfitting. However, this also prevented the baseline from learning excessively faster than the policy network, allowing the learning speeds between the baseline and the policy to be better aligned.

(b) **Policy performance**: When the quality of the baseline deteriorated, the accuracy of the advantage estimate (A_t) decreased, which made the policy gradient unstable. As a result, the Eval Average Return was lower than in the original baseline experiments, and the learning curve became slower with greater variability. This demonstrates that a good value function (baseline) is critical to the stability and performance of policy learning.



(a) Cheetah rollouts with n=100, b=5000 using NA

(b) Cheetah rollouts with n=200, b=10000 using NA

Figure 4: HalfCheetah-v4: Rollout with n=100, b=5000 (left) and Rollout with n=200, b=10000 (right) when using NA.

Although the Eval Return reached close to 300, the cheetah was unable to run properly and showed a tendency to fall over. This indicates that further training is necessary to achieve an Eval Return far exceeding 300 in order to demonstrate good performance when viewed in the actual video.

The Figure on the right shows how the cheetah moves when Eval Return exceeds 1000. We can see that it moves more stable

6 Generalized Advantage Estimator

6.1 Implementation

You will now use the value function you previously implemented to implement a simplified version of GAE-λ. This will require filling in the remaining TODO section in `PGAgent._estimate_advantage()`.

6.2 Experiments

Experiment 3 (HumanoidStandup-v5). You will now use your implementation of policy gradient with generalized advantage estimation to learn a controller for a version of `HumanoidStandup-v5`. Search over $\lambda \in [0, 0.95, 1]$ to replace $\langle\lambda\rangle$ below. Do not change any of the other hyperparameters (e.g. batch size, learning rate).

```
python cas4160/scripts/run_hw2.py \
    --env_name HumanoidStandup-v5 --ep_len 100 \
    --discount 0.99 -n 50 -l 3 -s 128 -b 2000 -lr 0.001 \
    --use_reward_to_go --use_baseline --gae_lambda <\lambda> \
    --exp_name HumanoidStandup_lambda<\lambda>
```

Deliverables:

- Provide a single plot with the learning curves for the `HumanoidStandup-v5` experiments that you tried. Describe in words how λ affected task performance and training efficiency.
- Consider the parameter λ. What does $\lambda = 0$ correspond to? What about $\lambda = 1$? Relate this to the task performance in `HumanoidStandup-v5` in one or two sentences.

What to Expect:

- The run with the best performance should achieve an average score close to $1.15e4$. In addition, each run takes about 10 minutes without video logging.

6.3 GAE Result

6.3.1 Plot (Eval Average Return with different λ):

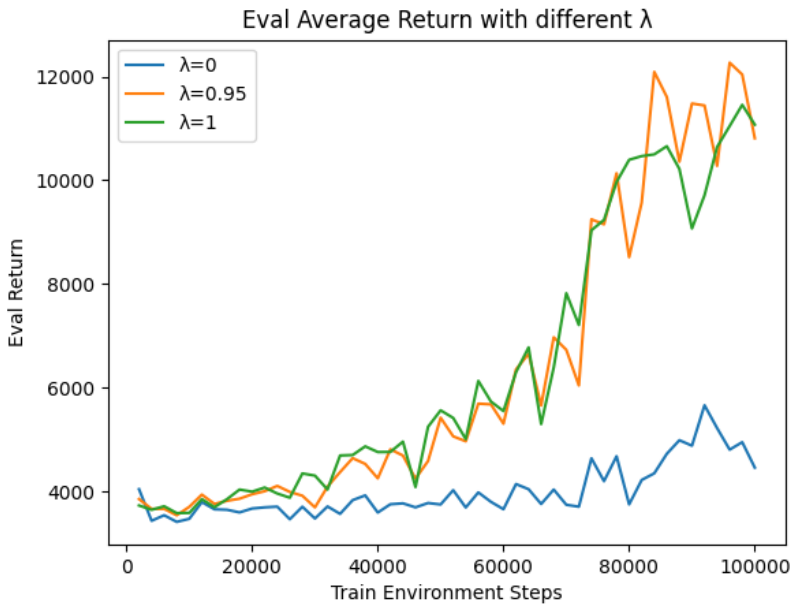


Figure 5: Eval Average Return curves for HumanoidStandup-v5 experiments with different GAE λ values ($\lambda = 0$, $\lambda = 0.95$, $\lambda = 1$).

6.3.2 Questions:

Answer the following questions briefly:

- Describe in words how λ affected task performance.
  Answer:

  As the value of λ increased, task performance improved significantly.

  When λ=0, learning was very unstable and the final performance was low.

  However, at λ=0.95, the learning curve showed good performance the fastest and recorded the highest Eval Average Return.

  When λ was increased to 1, the performance dropped very slightly again, and there was a tendency for a slight increase in learning variability. However, the difference compared to λ=0.95 was not as large as expected.

  Nevertheless, comparing λ=0 and λ=1, this result can be viewed as a clear demonstration of the bias-variance trade-off in GAE.

- Consider the parameter λ. What does $\lambda = 0$ correspond to? What about $\lambda = 1$? Relate this to the task performance in `HumanoidStandup-v5` in one or two sentences.
  Answer:

$\lambda = 0$ is the case that uses only the 1-step TD residual, which is an estimator with large bias but very low variance.

Conversely, $\lambda = 1$ is the case that uses the Monte-Carlo return as is, which has minimal bias but very high variance.

In environments like HumanoidStandup-v5 where the reward is sparse and the episode is long, $\lambda = 0.95$ achieved the optimal balance between bias and variance and showed the best performance, while $\lambda = 0$ was too conservative and $\lambda = 1$ was slightly unstable, resulting in both showing lower performance compared to $\lambda = 0.95$.

7 Proximal Policy Optimization

Finally, you are ready to implement Proximal Policy Optimization (PPO) algorithm. This algorithm uses generalized advantage estimates for calculating its surrogate advantage. There are two primary variants of PPO: PPO-Penalty and PPO-Clip. In this section, we will be implementing PPO-Clip, and its objective is defined by:

$$L\left(\mathbf{s}, \mathbf{a}, \theta_k, \theta\right) = \min\left(\frac{\pi_\theta(\mathbf{a} \mid \mathbf{s})}{\pi_{\theta_k}(\mathbf{a} \mid \mathbf{s})} A^{\pi_{\theta_k}}(\mathbf{s}, \mathbf{a}), \quad \text{clip}\left(\frac{\pi_\theta(\mathbf{a} \mid \mathbf{s})}{\pi_{\theta_k}(\mathbf{a} \mid \mathbf{s})}, 1-\epsilon, 1+\epsilon\right) A^{\pi_{\theta_k}}(\mathbf{s}, \mathbf{a})\right),$$

where θ_k refers to the old policy parameters before the update.

The main idea is that after an update, the new policy should be not too far from the old policy. For that, ppo uses clipping to avoid too large update. Having this form of regulation enables multiple epochs of minibatch updates with the same data, resulting in better stability and sample efficiency. If you want to know more about PPO, here is well-explained documents [OpenAI Spinning Up, PPO paper].

7.1 Implementation

To implement PPO, you need to fill in the TODO section in `MLPPolicyPG.ppo_update()` and `PGAgent.update()`. Please refer to the surrogate objective defined above for calculating the loss.

7.2 Experiments

Experiment 4 (Reacher-v4). Use your implementation of PPO with generalized advantage estimation to learn a controller for `Reacher-v4`. Run the following command to run the experiment.

```
# Reacher (Baseline)
python cas4160/scripts/run_hw2.py \
    --env_name Reacher-v4 --ep_len 1000 \
    --discount 0.99 -n 100 -b 5000 -lr 0.003 \
    -na --use_reward_to_go --use_baseline --gae_lambda 0.97 \
    --exp_name reacher
# Reacher (PPO)
python cas4160/scripts/run_hw2.py \
    --env_name Reacher-v4 --ep_len 1000 \
    --discount 0.99 -n 100 -b 5000 -lr 0.003 \
    -na --use_reward_to_go --use_baseline --gae_lambda 0.97 \
    --use_ppo --n_ppo_epochs 4 --n_ppo_minibatches 4 \
    --exp_name reacher_ppo
```

Deliverables:

- Provide a single plot with the learning curves for the Reacher-v4 experiments that you tried. Compare the performances of the final agent with and without using PPO.
- If we do not use surrogate objective and do multiple batch updates, what would happen? Justify the result.
- (Optional) You can verify the above question by changing "`actor.ppo_update()`" to "`actor.update()`" and comparing the performance.

What to Expect:

- The run with PPO should achieve an average score close to -10 if correctly implemented (at least -15). In addition, each run takes about 14 minutes without video logging.

7.3 PPO Result

7.3.1 Plot (Eval Average Return for PPO and the baseline):

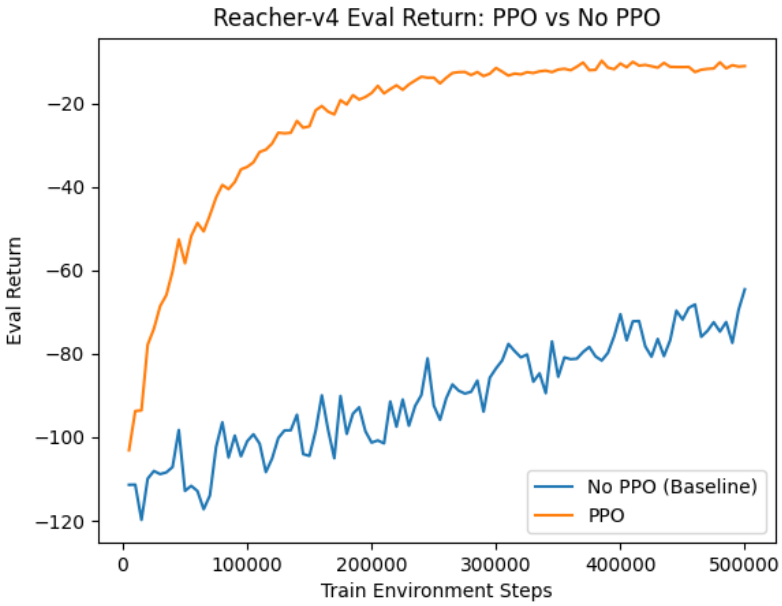


Figure 6: Eval Average Return curves for Reacher-v4 experiments. Comparison between the final agent with PPO and without PPO (Baseline PG + GAE).

In the Reacher-v4 experiment, we compared the performance when using PPO and when not using PPO (Baseline PG + GAE) in Figure 6.

- **When PPO was used**: Learning was very stable, and the Eval Average Return improved rapidly, eventually converging to approximately $-10 \sim -11$. Thanks to the clipping mechanism, policy updates did not become excessively large, so policy collapse almost never occurred, and the same data could be safely reused across multiple epochs.

- **When PPO was not used (Baseline PG + GAE)**: The learning curve was unstable and exhibited very high variability, with the final Eval Average Return remaining at the level of $-60 \sim -80$. When the advantage was large, the policy was updated significantly in a single step, and a sharp drop in reward in the next rollout was frequently observed.

Overall, the agent using PPO outperformed the No-PPO agent in terms of learning speed, stability, and final performance. This is the result of PPO's surrogate objective and clipping effectively limiting the magnitude of policy updates, thereby preventing policy collapse and improving sample efficiency.

7.3.2 Plot (Reacher-v4: Effect of removing Surrogate Objective)

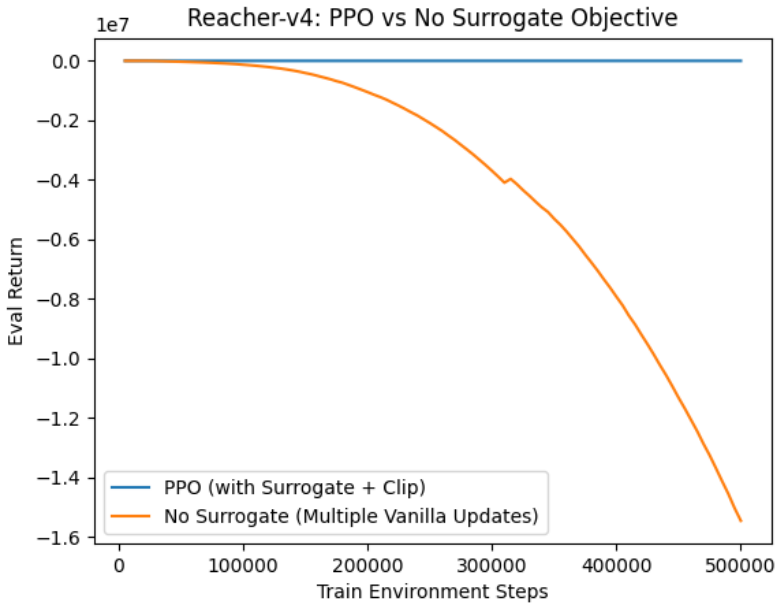


Figure 7: Eval Average Return curves for Reacher-v4 when using PPO (with surrogate objective + clipping) versus using multiple vanilla updates without surrogate objective.

7.3.3 Questions:

Answer the following questions briefly:

- If we do not use surrogate objective and do multiple batch updates, what would happen? Justify the result.
  Answer:

  Without using the surrogate objective, training the same rollout data multiple times (`n_ppo_epochs=4`) with vanilla updates (`actor.update()`) leads to policy collapse, making the learning extremely unstable and significantly degrading performance.

  Specifically, the following phenomena were observed:

  - At timesteps where the advantage (A_t) is large, the policy ratio $r_t(\theta) = \frac{\pi_\theta(a|s)}{\pi_{\theta_{\text{old}}}(a|s)}$ deviates significantly from 1, causing the policy to be updated excessively in a single step.

- As a result, in the next rollout, the policy performs completely different actions compared to before, leading to a sharp drop in reward.
- Consequently, the learning curve became highly unstable, and the Eval Average Return converged to a significantly lower value than when PPO was used.

As can be seen in Figure 7, PPO using the surrogate objective + clipping was able to safely reuse the data over multiple epochs by limiting the magnitude of policy updates to the range $1 - \epsilon \sim 1 + \epsilon$. In contrast, when the surrogate objective was removed, repeatedly updating the same data caused the policy to become increasingly unstable, resulting in a substantial drop in final performance.

This clearly demonstrates that PPO's surrogate objective (= clipping) is the core mechanism that prevents the policy from deviating too far from the old policy. Without the surrogate objective, multiple batch updates actually hinder learning rather than helping it.

8 Submission

Please include the code, tensorboard logs data, and the **"report"**. Zip it to `hw2_[YourStudentID].zip`. The structure of the submission file should be:

```
hw2_[YourStudentID].zip
├── hw2_[YourStudentID].pdf
├── cas4160/
│   └── ...codes
└── ...
```

Note: Do NOT include the videos (.mp4 files or tensorboard logs) in your submission. Your submission file size should be less than 50MB.